Parv **Kapoor**

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Education

Carnegie Mellon University — *Pittsburgh, PA*
Ph.D. in Software Engineering | Advisors: Dr. Eunsuk Kang and Dr. Sebastian Scherer — *May 2026 (expected)*

Carnegie Mellon University — *Pittsburgh, PA*
Masters in Software Engineering | TA: 10-623 Generative AI, 17-723 Designing large scale systems — *December 2025 (expected)*

Manipal Institute of Technology — *Manipal, India*
B.Tech. in Computer Science Engineering — *August 2020*

Thesis: Specification-Driven Foundation Models for Safe Autonomous Systems

Skills

Programming	Python, C, C++, CUDA programming, Alloy
Tools and Libraries	PyTorch, JAX, ROS, Docker, PyTorch Lightning, Isaac Sim

Ph.D. Research

AirLab, *Carnegie Mellon University* — *Pittsburgh, PA*
PI: Sebastian Scherer — *August 2021 – Ongoing*

- SafeDec: Developed constrained decoding frameworks enabling safety enforcement for robot foundation models without retraining
- STLCG++ [RA-L '26]: Built a PyTorch/JAX library for differentiable logic, enabling 100× faster robustness computation via adaptive masking
- ViSafe [RSS'25]: Developed vision-based control barrier functions for UAV safety, validated via sim-to-real transfer from Isaac Sim to field, achieving 71% improvement over baselines across 70+ flight hours
- MCSTL [ICRA '23]: Trained real-world aviation policies using guided imitation learning and online tree search, outperforming baselines by 60%

Software Design and Analysis Lab, *Carnegie Mellon University* — *Pittsburgh, PA*
PI: Eunsuk Kang — *August 2021 – Ongoing*

- ETL: Trained world models and used pretrained visual embeddings (DINOv2, CLIP) with MPPI for logic-constrained object navigation
- ATLAS [ICSE'25]: Learned formal behavior rules from robot demonstrations using MaxSAT for structured policy synthesis
- RobRL [FM'24]: Designed stochastic optimization methods to analyze reinforcement learning policy robustness under deviations
- STLINC [NFM'24]: Devised requirement decomposition for integrated task and motion planning, reducing solve time by 65%

Experience

Microsoft Research — *New York, NY*
Machine Learning Research Intern | PI: Dr. John Langford and Dr. Siddhartha Sen — *May 2025 – August 2025*

- Built Matryoshka Belief State Transformer (MatBST) architecture with hierarchical latent planning, enabling fast long-horizon reasoning in LLMs
- Optimized large-scale pretraining and fine tuning with PyTorch FSDP and custom sharding strategies for large language models

General Robotics — *Seattle, WA*
Student Researcher | PIs: Dr. Jonathan Huang and Dr. Ashish Kapoor — *September 2024 - May 2025*

- Developed pipelines for procedural synthetic data generation for AI2-THOR, Habitat and Unreal Engine environments
- Pretrained multi-modal robot foundation models via large-scale behavior cloning for SOTA mobile manipulation
- Developed aligned robotics foundation models using parameter-efficient and reinforcement fine-tuning for cross-task generalization

General Robotics — *Seattle, WA*
Research Intern | PIs: Dr. Sai Vemprala and Dr. Ashish Kapoor — *May 2024 - August 24*

- PASTEL (RSS '24): Trained instruction-conditioned robot policies with cross-attention transformers, outperforming baselines by 74.3%
- Contributing to the development of the AirGen simulation platform for aerial and ground robots
- Built a foundation model field deployment pipeline for diverse robots (Unitree Go2, DJI Robomaster) using domain adaptation

Cyber Physical Systems Lab, *University of Southern California* — *Los Angeles, U.S.A.*
Research Intern | PI: Dr. Jyotirmoy Vinay Deshmukh — *January 2020 - January 2021*

- Developed novel model-based reinforcement learning algorithms for safe policy training from signal temporal logic specifications
- Implemented efficient model-free algorithms (TRPO, A3C, PPO) in PyTorch with unique STL-based reward design
- Achieved 82% higher specification satisfaction compared to baseline RL policies
- Engineered in-house simulation environments for algorithm benchmarking employing CARLA, AirSim, and Gazebo

Visual Computing Group, *Cardiff University* *Cardiff, U.K.*

Research Intern | PI: Dr. David Marshall *May 2019 - July 2019*

- Constructed a safe trajectory prediction system for visually impaired individuals using ZED stereo camera
- Implemented and trained a 2-stream CNN in TensorFlow on human walking data for forecasting ego agent camera movement
- Improved CNN accuracy in low-data regimes through neuro-inspired data augmentation

Preprints and Publications

Scaling Long-Horizon Reasoning via Hierarchical Matryoshka Belief State Transformers

P. Kapoor, S. Sen, J. Langford

- Under Submission

Constrained Decoding for Robotics Foundation Models

P. Kapoor, A. Ganlath, C. Liu, S. Scherer, E. Kang

- Under Submission [arxiv]

Pretrained Embeddings as a Behavior Specification Mechanism

P. Kapoor, A. Hammer, A. Kapoor, K. Leung, E. Kang

- Under Submission [arxiv]

ViSafe: Vision-enabled Safety for High-speed Detect and Avoid

P. Kapoor, I. Higgins, N. V. Keetha, J. Patrikar I. Cisneros, Z. Ye, Y. He, Y. Hu, C. Liu, E. Kang, S. Scherer

- **Robotics Science and Systems (RSS)** 2025 [arxiv]

STLCG++: A Masking Approach for Differentiable Signal Temporal Logic Specification

P. Kapoor, K. Mizuta, E. Kang, K. Leung

- **IEEE Robotics and Automation Letters (RA-L)**, presented at IEEE International Conference on Robotics and Automation (ICRA) 2026 [arxiv]

Logically Constrained Robotics Transformers for Enhanced Perception Action Planning

P. Kapoor, S. Vemprala, A. Kapoor

- **Robotics Science and Systems (RSS): Towards Safe Autonomy** 2024 [arxiv]

Constrained LTL Specification Learning from Examples

C. Zhang*, **P. Kapoor***, I. Dardik, A. Cui, R.M. Goes, D. Garlan, E. Kang

- **IEEE International Conference on Software Engineering (ICSE)** 2025 [arxiv]

Analyzing Tolerance of Reinforcement Learning Controllers against Deviations in Cyber Physical Systems

C. Zhang*, **P. Kapoor***, R.M. Goes, D. Garlan, E. Kang, A. Ganlath, S. Mishra, N. Ammar

- **Formal Methods (FM)** 2024 [arxiv]

Safe Planning through Incremental Decomposition of Signal Temporal Logic

P. Kapoor, R.M. Goes and E. Kang

- **Nasa Formal Methods (NFM)** 2024 [arxiv]

Follow The Rules: Online Signal Temporal Logic Tree Search for Guided Imitation Learning in Stochastic Domains

J. Patrikar, J. Aloor, **P. Kapoor**, S. Scherer and J. Oh

- **IEEE International Conference on Robotics and Automation (ICRA)** 2023 [arxiv]

Challenges in Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace

J. Patrikar, J. Dantas, S. Ghosh, **P. Kapoor** et al

- **IEEE International Conference on Robotics and Automation (ICRA)** 2022 [arxiv]

Model-based Reinforcement Learning from Signal Temporal Logic Specifications

P. Kapoor, A. Balakrishnan, J. V. Deshmukh

- [arxiv]